

Simulation of Aircraft Manoeuvres Based On Computational Fluid Dynamics

M. Ghoreyshi*, D. Vallespin †, A. Da Ronch ‡ and K. J. Badcock§

Department of Engineering, University of Liverpool, Liverpool, UK, L69 3GH

J. Vos ¶

CFS Engineering, Lausanne, Switzerland

S. Hitzel||

EADS Military Air Systems, Munich, Germany

The use of computational fluid dynamics to generate and test aerodynamic data tables for flight dynamics analysis is described in this paper. The test case used is the Ranger 2000 fighter trainer for which flight test data is available. The generation of the tables is done using sampling and reconstruction to allow a large number of table entries to be generated at low computational cost. The testing of the tables is done by replaying, through a time accurate CFD calculation which features the moving control surfaces, manoeuvres and comparing the forces and moments against the tabular values. The manoeuvres are generated using a time optimal prediction code with the feasible solutions based on the tabular aerodynamics. The generated manoeuvres are evaluated against flight data to show that they are qualitatively representative. Then the time accurate and tabular aerodynamics are compared, and as expected are in close agreement.

I. Introduction

The aerodynamic forces acting on aircraft need to be estimated for the analysis of the vehicle's motion. The formulation of these forces date back to the early 1900s, when G.H. Bryan and W.E. Williams¹ introduced the first stability-derivatives aerodynamic model, which is still found reliable for conventional aircraft in aerodynamically benign regions of the flight envelope. To meet the demands of accurately modeling the aerodynamics of manoeuvring aircraft, some improved methods are required for representing the nonlinear and unsteady flows around the aircraft.²⁻⁴ The source of nonlinearity and unsteadiness is mainly due to shock waves, separation and vortices at high angles of attack.⁵ The prediction of the flows over aircraft with high angle of attack and large amplitude manoeuvres is complicated by the fact that aerodynamic forces and moments of aircraft responding to sudden changes of flow not only depends on the instantaneous values but also the time history of the motion.⁶ There have only been a limited number of attempts to include the nonlinearity and unsteadiness. Examples include the work of Tobak and his colleagues⁷ for the non-linear indicial response methods, and Goman and Khrabov for the state-space model.⁸ The basic parameters of both methods are often identified based on the measured flight test data,⁹ but for a comprehensive model, a large number of training manoeuvres are needed that makes such a model very expensive.

Traditionally, aerodynamic data are obtained from wind tunnel testing or flight test data. The wind tunnel experiments can be expensive and suffer from scaling issues, along with difficulty in representing some dynamic motions.^{10, 11} The flight test data are also very expensive and available late in the development. Today, CFD tools are becoming credible for simulating the physics of the aerodynamic flow. In principle,

*Research Associate, AIAA member

†Ph.D Student, AIAA member

‡Ph.D Student, AIAA member

§Professor, AIAA senior member, Corresponding author, E-mail: k.j.badcock@liverpool.ac.uk

¶Director of CFS Engineering

||Technical Head and Doctor of Aerodynamics

CFD can produce the aerodynamic inputs for the formulation of aircraft loads, also, simulate directly the aerodynamic responses of a manoeuvring aircraft by coupling the time-dependent Reynolds-Averaged Navier-Stokes (RANS) equations and the dynamic equations governing the aircraft motion. The moving rigid body aerodynamics has been studied over the years, but its use was rather limited to the prediction of aircraft dynamic derivatives^{12,13} or two-dimensional test cases.^{14–17} Recently, there is an increasing interest of the coupled CFD-flight dynamics of a full geometry.^{18–20} Although the coupling is not likely to be a tool for routine flight mechanics studies due to expensive computational cost, such a simulation could be very powerful for investigating and understanding potential problem with the aerodynamic models.

The aerodynamic model considered in this paper is tabular in form. Tables, in contrast to stability derivatives, are not linearized, and are consistent with quasi-steady aerodynamics for a wide range of flight conditions. There are several difficulties associated with aerodynamic tables. First, a large number of table entries must be filled. The brute-force calculation of the entire table using CFD is not feasible due to computational cost. Sampling and data fusion methods have been proposed to overcome this.²¹ Secondly, the pre-computed nature of tables lacks the ability to describe hysteresis of the aerodynamic phenomena.

The current study uses access to flight test data for the Ranger 2000 to evaluate CFD generated aerodynamics and manoeuvres. The four objectives of this paper are: (a) demonstrate the sampling/Kriging approach to generating aerodynamic data tables; (b) test the realism of the manoeuvres generating using an optimisation procedure; (c) compare the CFD predicted aerodynamics with the identified model; (d) evaluate tables using time accurate replay. The paper first reviews the flow solver and how the manoeuvres are computed. Then the test case is described and the aerodynamic predictions validated. The generation of the aerodynamic tables and the approach to defining the time optimal manoeuvres is then given. The evaluation of the aerodynamic tables for these manoeuvres is then made by replaying them directly through an unsteady CFD calculation. Finally, conclusions are given.

II. Formulation

A. CFD

The flow solver used for this study is the University of Liverpool PMB (Parallel Multi-Block) solver. The Euler and RANS equations are discretised on curvilinear multi-block body conforming grids using a cell-centred finite volume method which converts the partial differential equations into a set of ordinary differential equations. The equations are solved on block structured grids using an implicit solver. A wide variety of unsteady flow problems, including aeroelasticity, cavity flows, aerospike flows, delta wing aerodynamics, rotorcraft problems and transonic buffet have been studied using this code. More details on the flow solver can be found in Badcock et al²² and a validation against flight data for the F-16XL aircraft is made in reference.²³

B. Table Generation

The aerodynamic model considered in this paper is tabular in form. The model consists of tables of forces and moments for a set of aircraft states and controls (e.g. aircraft angle of attack, side-slip angle, control deflections, etc.), which spans the flight envelope. This potentially entails a large number of calculations, which will be a particular problem due to the computational cost if CFD is the source of the data. This issue has been addressed by sampling and reconstruction based on Kriging interpolation and data fusion using Co-Kriging as described in reference.²¹

Two scenarios were considered, based on (1) a requirement for the generation of static table that include the table of Mach number and angles of attack and side-slip and (2) for updating the static table for control tables of elevator, rudder and aileron. Both approaches are available in the Computerized Environment for Aircraft Synthesis and Integrated Optimization Methods, CEASIOM,²⁴ computer code.

In the first scenario it is assumed that the static table is generated without user intervention. The emphasis is on a sampling method which identifies the nonlinearities in the force and moment tables with respect to the static aircraft state parameters. Approaches to the sampling based on the Mean Squared Error (MSE) criterion of Kriging and the Expected Improvement Function (EIF) were considered in reference.²¹

In the second scenario, it is assumed that the control tables are incremented from the static table, i.e. the static table represents the main trends of aerodynamic forces and moments, while the controls result in the increments of these trends. Data fusion based on Co-Kriging is then used to update the static table,

based on a small number of calculations of control surface deflections.

Using these techniques it was shown that tables which are practically useful could be generated in the order of 60 calculations under the first scenario and 10 calculations under the second scenario.

C. Time Optimal Manoeuvres

We use the optimal control approach,^{25,26} that finds the optimal controls that transfer a system from the initial state to the final state while minimizing (or maximizing) a specified cost function.²⁷ The optimal control aims to find a state-control pair $x^*(\cdot), u^*(\cdot)$ and possibly the final event time t_f that minimizes the cost function

$$J[x(\cdot), u(\cdot), t_0, t_f] = E(x(t_0), u(t_f), t_0, t_f) + \int_{t_0}^{t_f} F(x(t), u(t), t) dt \quad (1)$$

where $x \in R^n$ and $u \in R^m$. The function E and F are endpoint cost and Lagrangian (running cost), respectively.

There are many different methods of solving the optimal control problems, in the current paper, the commercially available code, DIDO²⁸ and MATLAB²⁹ are used. In DIDO, the total time history is divided into N segments, spaced using shifted Legendre-Gauss-Lobatto (LGL) rule.³⁰⁻³² The boundaries of each time segment are called nodes. the code exploits pseudo-spectral (PS) methods for solving the optimal control problems.

For the problem of an aircraft optimal time manoeuvre, the general 6-degree-of-freedom aircraft equations of motion detailed in Etkin³³ and Stevens and Lewis³⁴ serve as one of the constraints. The aircraft state vector consists of the position of the aircraft (x, y, z), the standard Euler angles (ϕ, θ, ψ), the velocity components in terms of Mach number and flow angles (M, α, β), and the body-axis components of the angular velocity vector (p, q, r). The initial and final state parameters are fixed with trimmed flight conditions, but the rest of the manoeuvre is out of trim conditions. The loads in the aircraft equations of motion are interpolated from the generated look-up tables.

D. Replay

The key functionality for the CFD solver in the current application is the ability to move the mesh. Two types of mesh movement are required. First, a rigid rotation and translation is required to follow the motion of the aircraft. Secondly, the control surfaces are deflecting throughout the motion. The control surfaces are blended into the geometry in the current work following the approach given in reference.³⁵ After the surface grid point deflections are specified, transfinite interpolation is used to distribute these deflections to the volume grid.

The rigid motion and the control deflections are both specified from a motion input file. For the rigid motion the location of a reference point on the aircraft is specified at each time step. In addition the rotation about this reference point is also defined. Mode shapes are defined for the control surface deflections.¹⁸ Each mode shape specifies the displacement of the grid points on the aircraft surface for a particular control surface. These are prepared as a preprocessing step using a utility that identifies the points on a control surface, defines the hinge, rotates the points about the hinge and works out their displacements. The motion input file then defines a scaling factor for each mode shape to achieve the desired control surface rotation.

The desired motion to be replayed through the unsteady CFD solution is specified in the motion input file. The aircraft reference point location, rotation angles and control surface scaling factors are needed. The rotation angles are obtained straight from the pitch, yaw and bank angles. The aircraft reference point velocity $\mathbf{v}_a$ is then calculated to achieve the required angles of attack and sideslip, and the forward speed. The velocity is then used to calculate the location. The CFD solver was originally written for steady external aerodynamics applications. The wind direction and Mach number are specified for a steady case. If the initial aircraft velocity is denoted as $\mathbf{v}_0$, then this vector is used to define the Mach number and the angle of attack for the steady input file. Then the instantaneous aircraft location for the motion file is defined from the relative velocity vector $\mathbf{v}_a - \mathbf{v}_0$.

III. Test Case

The aircraft considered in this paper is the Ranger 2000. This is a mid-wing, tandem seat training aircraft powered by one turbofan engine with uninstalled thrust of 14190 N. The wing and fuselage are manufactured

of composite material and the empennage is a metal T-tail design. A three-view of the vehicle is shown in Fig. 1. Also, the general dimensions and the mass/inertial properties for both maximum take-off weight (MTOW) and operating empty weight(OEW) are listed in tables 1 and 2.

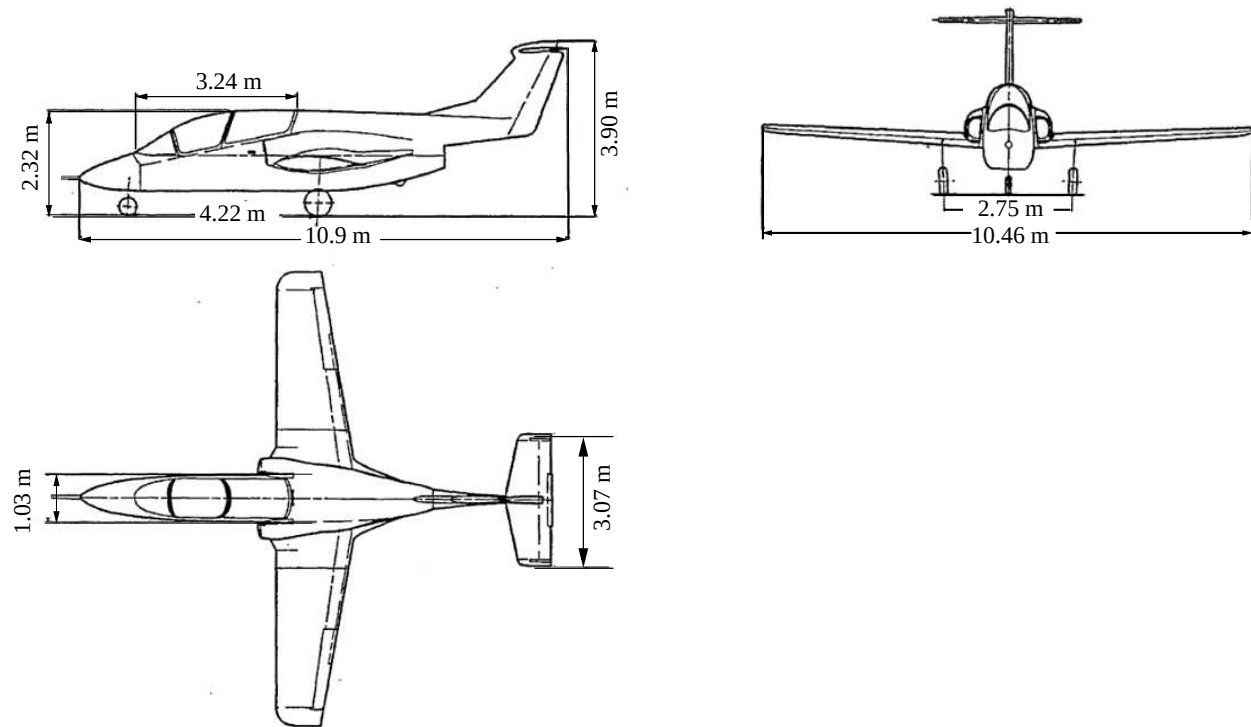


Figure 1. Three-Views of Ranger 2000

Table 1. General Dimensions of Ranger 2000

Length Overall (m)	10.39
Wing Span (m)	10.46
Height Overall (m)	3.58
Wing Area m ²	15.5
Mean Aerodynamic Chord (MAC) (m)	1.545
Wing Taper Ratio	0.519

The operational envelope in terms of the flight altitude and speed is shown in Fig. 2 for the specified weight of 3765kg. The envelope has a maximum operational ceiling of 9449 m and the diving speed (M_D) and the maximum operating speed (M_{MO}) of 0.7 and 0.75, respectively.

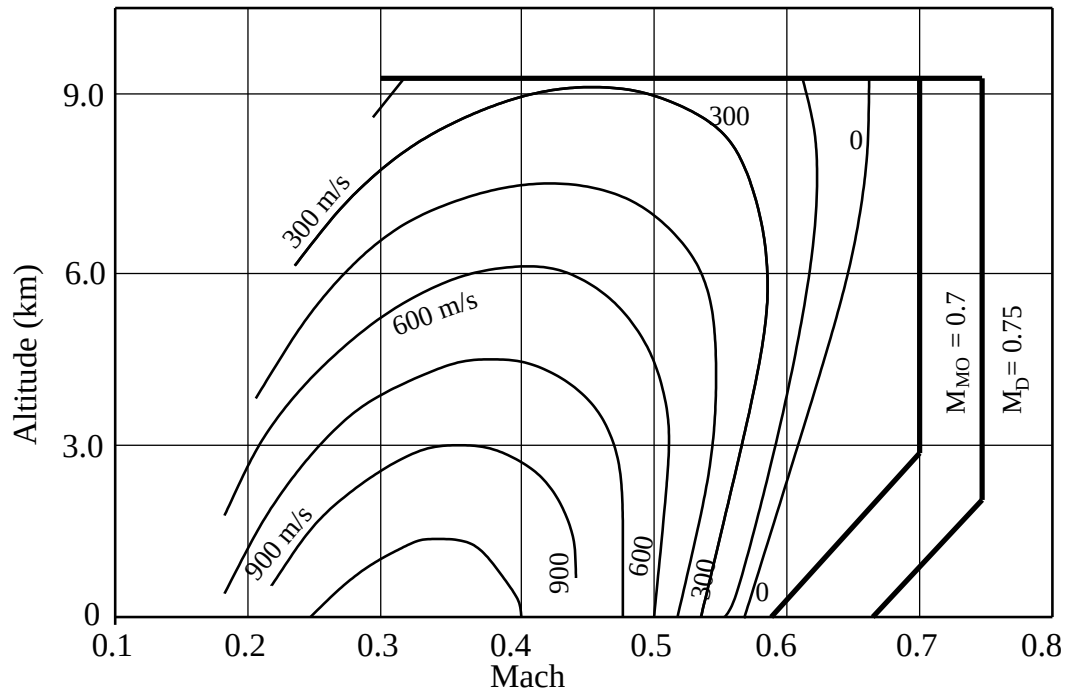
The vehicle flight control system consists of three conventional control surfaces: The elevator at the tail, the rudder at the fin and left and right ailerons (Fig. 3). The position limits of each surface is given in table 3, with positive angles indicating the elevator trailing edge down, the left aileron trailing edge down and the rudder trailing edge toward the left wing (see Fig. 3).

The flight test data consists of all aerodynamic forces and moments with respect to the aircraft states of: angle of attack, side-slip angle, Mach number, rotational rates, acceleration rates, elevator, rudder, control and the altitude of flight. Table 4 summarizes the range of measured data. Various aerobatic manoeuvres were performed to demonstrate general aircraft handling qualities. These include Barrel Rolls, Clover Leafs, Immelmann Turns, inverted flight, Lazy Eights, Loops, and Split-S. The entry conditions and the time histories for each manoeuvre are provided by EADS military air systems.

A multiblock structured grid was generated using the commercial grid generation tool ICEMCFD. Both

Table 2. Mass/Inertias of Ranger 2000

	MTOW	OEW
Ixx (kgm ²)	9287.1	4600.9
Iyy (kgm ²)	13584.1	13462.3
Izz (kgm ²)	21237.6	16474.6
MTWO (kg)	2586	
MTWO(maximum) (kg)	3765	

**Figure 2. Ranger 2000 Accelerated Climb Rate with 3765 kg Gross Mass****Table 3. Control surface deflection limits.**

Elevator	δ_{ele}	$-25/+15^0$
Right Aileron	δ_{ail}	$-25/+15^0$
Left Aileron	δ_{ail}	$-25/+15^0$
Rudder	δ_{rud}	$\pm 17.5^0$

Table 4. The boundary of experimental data

Angle of Attack	α	$-7/+36^0$
Side-slip Angle	β	$-20/+20^0$
Mach number	M	$0/0.75$
Altitude	h	$0/10$ km

full and half configurations were generated, the latter to save on computing costs in the simulation of longitudinal flight dynamics. The lateral flight dynamics tables, on the other hand, required the full model flow prediction. The half configuration has 14.5 million points arranged in 2028 blocks.

Figure 4 (a) shows the overall view of the meshed geometry. Figures 4 (b) and (c) show the diamond shaped tip topology used to accomodate the cells around the main wing and tail-plane. The wing has a H-type topology around the leading edge to improve the cell quality in the wing-engine-fuselage junction while the horizontal tail-plane blocking consists of a C-type round its LE.

The aforementioned control surfaces are also included in the geometry (Fig. 3). These were the elevator, aileron and rudder which can be deflected for steady state calculations and during time accurate simulations. The half grid requires 1.5 hours on 128 processors on the United Kingdom's academic supercomputer (Cray inc) to obtain a fully converged steady state solution.

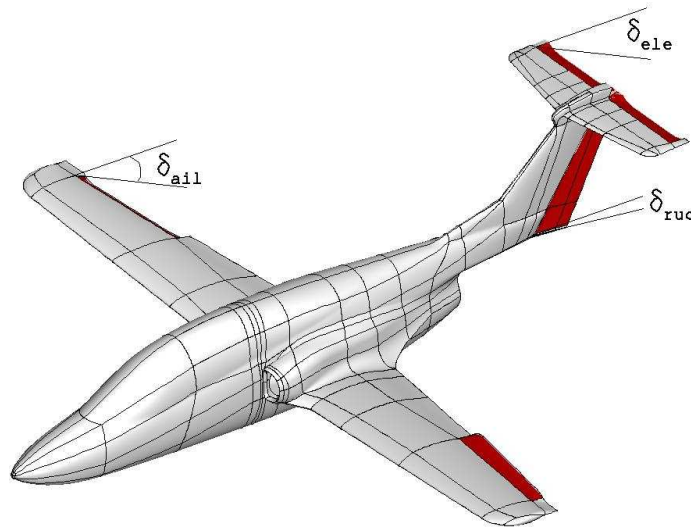


Figure 3. Ranger control surfaces- The configuration shows the positive deflection angles.

IV. Results

A. Generation of Aerodynamic Tables

The format of the aero look-up table used in the CEASIOM code is given in table 5. The aerodata provides the aerodynamic forces and moments for each point corresponding to the aircraft states.

Table 5. CEASIOM aerodata model

α	M	β	δ_{ele}	δ_{ail}	δ_{rud}	C_L	C_D	C_m	C_Y	C_l	C_n
x	x	x	-	-	-	x	x	x	x	x	x
x	x	-	x	-	-	x	x	x	x	x	x
x	x	-	-	x	-	x	x	x	x	x	x
x	x	-	-	-	x	x	x	x	x	x	x

where α is the angle of attack, M is the Mach number and β is the side slip angle. The last six columns are the coefficients of lift, drag, pitching-moment, side-force, rolling moment and yawing moment. Based on this format, four three-parameter sub-tables need to be generated.

The table (angle of attack, Mach number and side-slip angle) was generated from scratch using sampling. A 500 entry table was first generated using the PMB solver for 65 calculations. The generated table describes the underlying behaviour of aerodynamic forces and moments. Note that the angles of attack and Mach number in the tables are limited to 12 degrees and 0.6, respectively. All flight test data are within these limits.

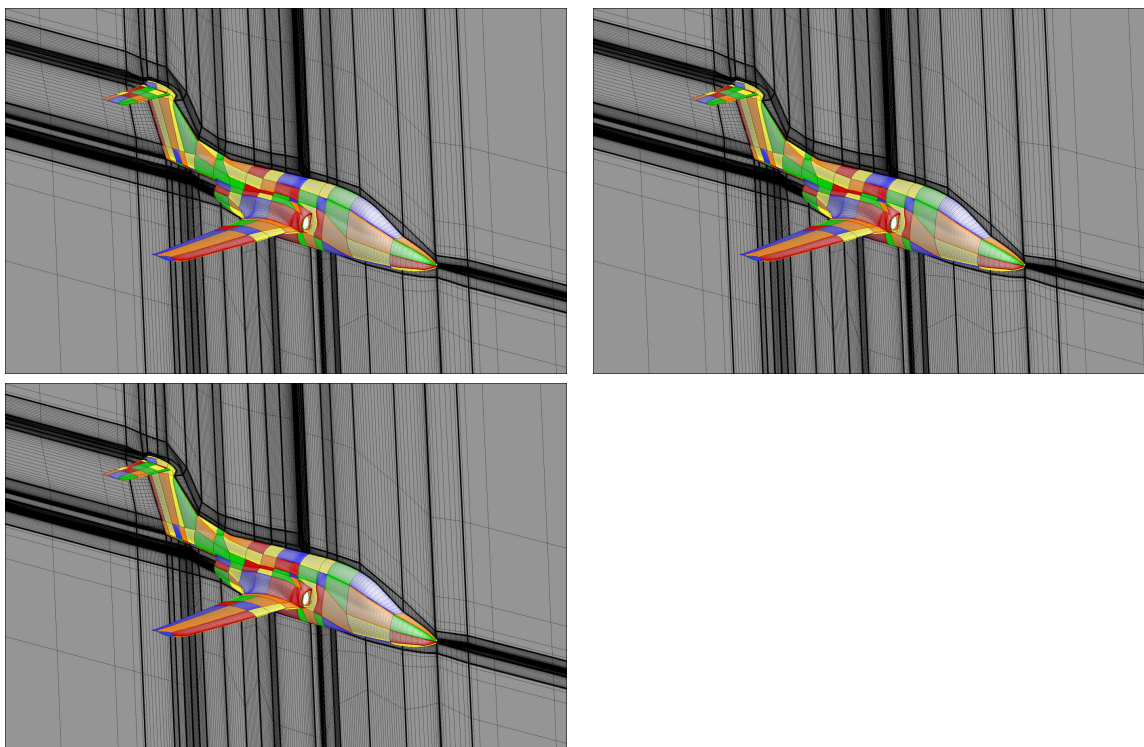


Figure 4. The Ranger grid

Next, the tables for varying angle of attack, Mach number and control surface deflections were generated. Twelve additional samples for each configuration at non-zero deflection angles were calculated. The values at these samples were then used to increment the static table values using Co-Kriging. For the purpose of generation of manoeuvres, instead of the actual values, the increments of aerodynamic forces and moments are stored in tables, i.e. $(\Delta C_{L\delta_{ele}}, \Delta C_{D\delta_{ele}}$ and etc).

Wind tunnel and flight data for the aerodynamic coefficients are available at low and transonic speeds. The comparisons between the measurements and Euler predictions using PMB are shown in Figs 5-6 for two speeds. The lift coefficient results match the measurements well, with the exception of an overprediction at high angle of attack. At low speeds, the drag is shifted down when compared with the measurements, reflecting the lack of the skin friction contribution. Also, the pitching moment curve slope from the Euler results is more negative.

The drag predictions at transonic speed become closer as the angle of attack increases, reflecting the increasing dominance of wave drag as the shock wave increases in strength. The formed shock on the upper surface of the wing at eight degree angle of attack is presented in Fig. 7. As the angle of attack increases at transonic speed, the shock becomes stronger and this results in a forward shift of the aerodynamic centre and consequently changes the slope of pitching moment (see the pitching-moment curve in Fig. 6). Inspecting the CFD solutions shows that the Euler solver does predict the shock formation and movement. The Euler solutions also predict the trends of the lateral coefficients, however, the results do not match the measurements due to the lack of the skin friction contribution.

Wind tunnel data for the aerodynamic coefficients are available at low speed. The trend of control power with angle of attack from CFD and experimental data is compared in Figs. 8(c)-(e). The flight speed is $M=0.25$ with zero side-slip angle for all the figures. The results of two elevator surface deflections are presented, while the angle of deflection for rudder and aileron is $+9$ degree. The results are in agreement with measured data in particular at low angles of surface deflection. Difference are possibly due to using the blended approach for the treatment of surface deflections.

In this paper, the dynamic effects are modeled by the stability derivatives that represent the influence of the aircraft rates around the three body axes on the aerodynamic forces and moments. A well-established

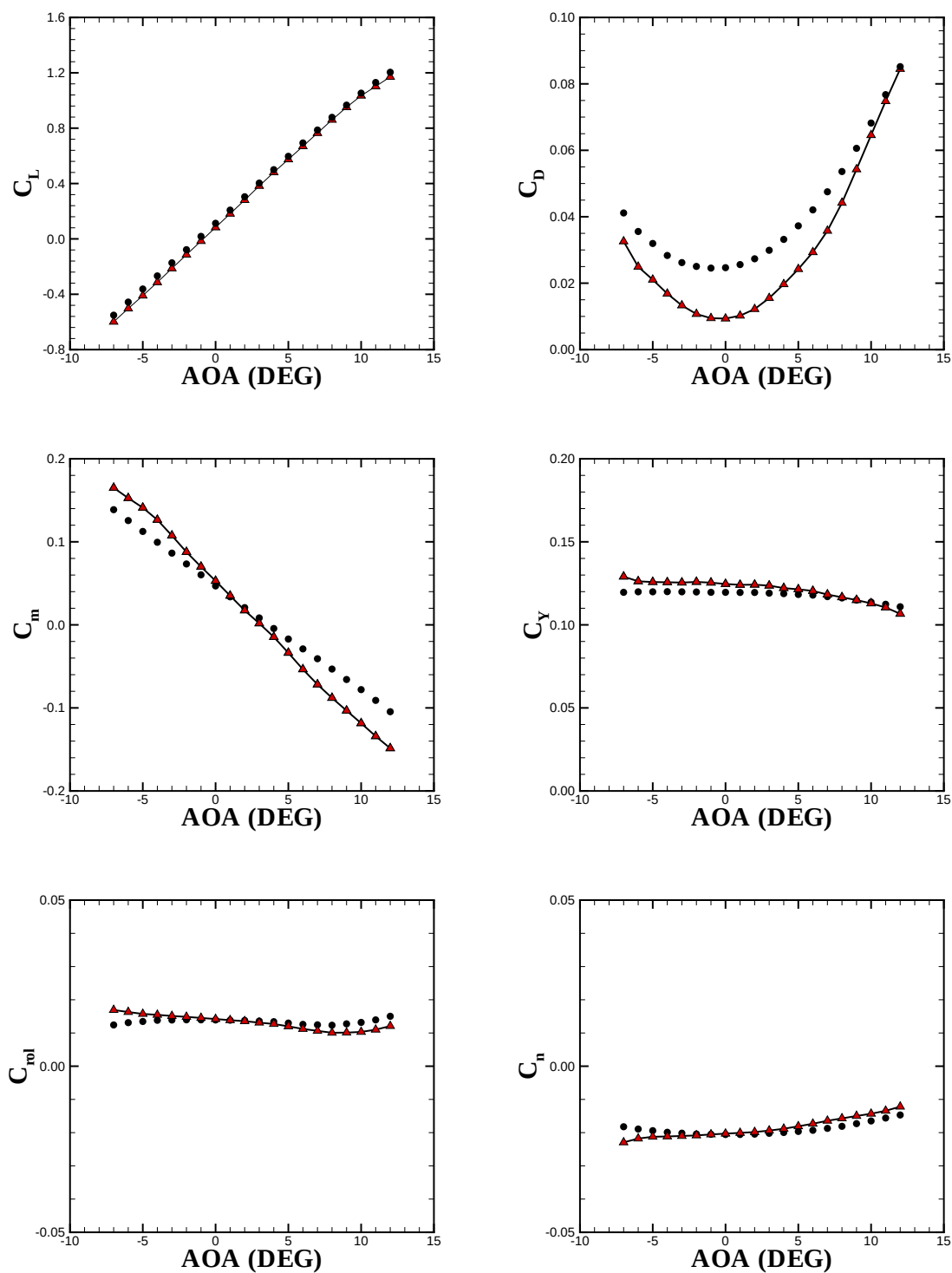


Figure 5. Validation of static table at $M=0.25$. The lift, drag and pitching moment correspond to zero side-slip angle, while the side-slip angle in lateral coefficients have is -8° . The circles denote experiments. All moment coefficients are in the body axis

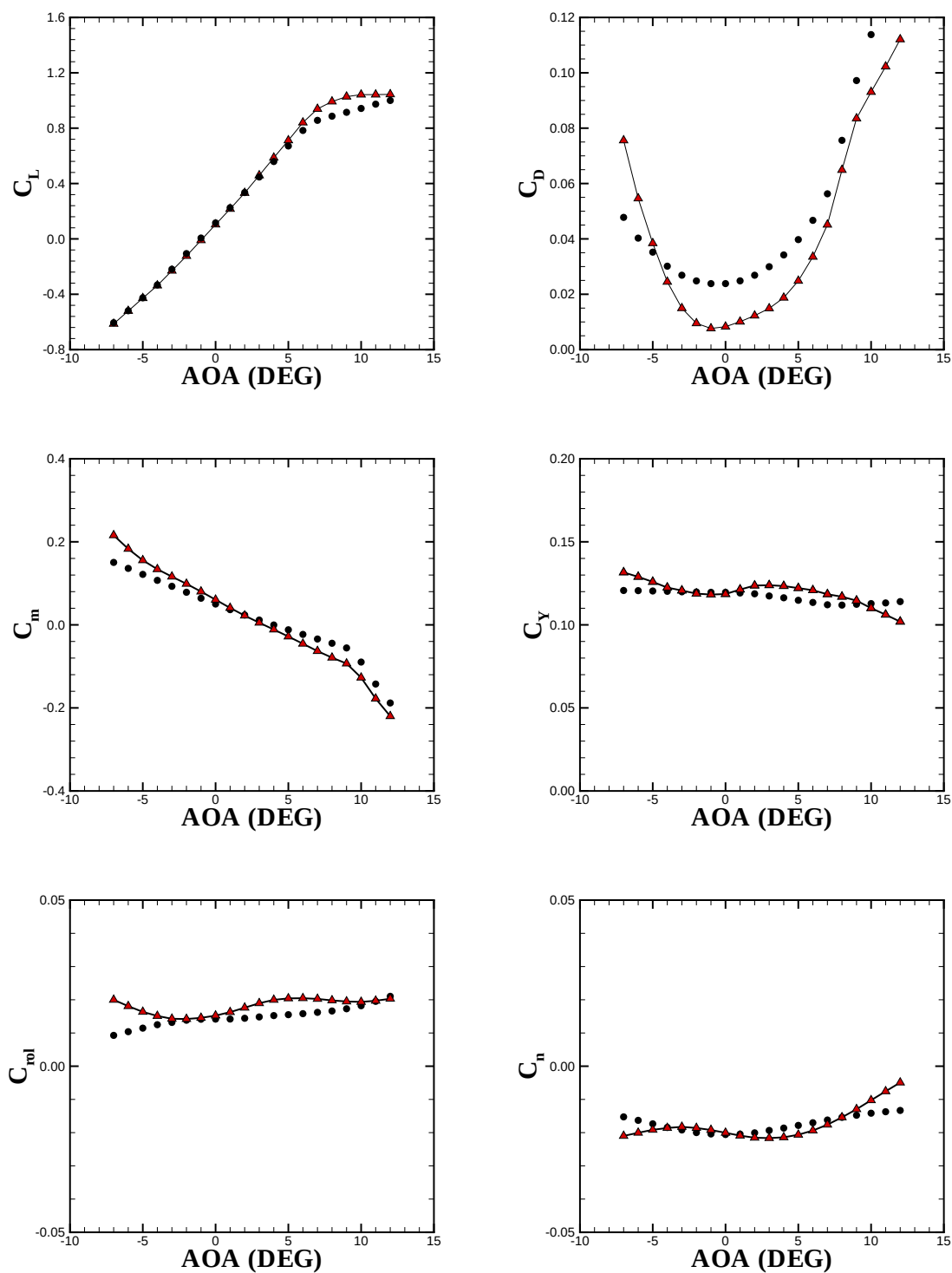


Figure 6. Validation of static table at $M=0.6$. The lift, drag and pitching moment correspond to zero side-slip angle, while the side-slip angle in lateral coefficients have is -8° . The circles denote experiments. All moment coefficients are in the body axis

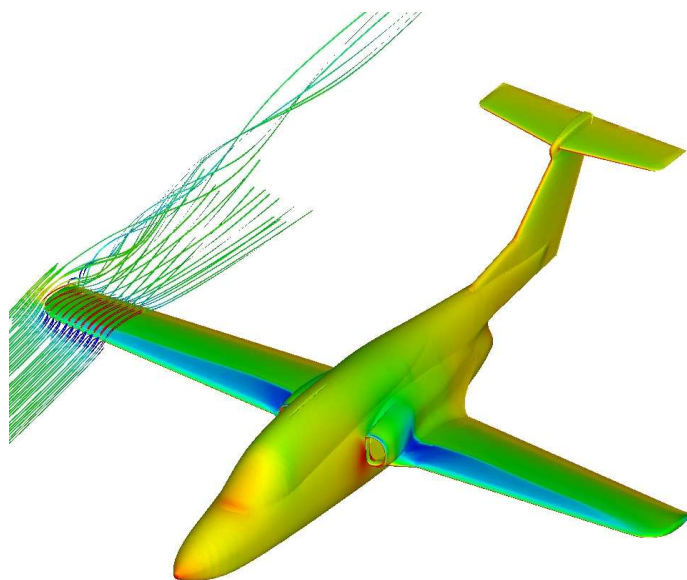


Figure 7. Surface solution at $\text{AOA} = 8$ and $M=0.6$

framework¹² for the prediction of dynamic derivatives was defined using the PMB solver. The dynamic derivatives were estimated by processing the solution in the time domain using a developed FFT algorithm. A linear regression model was also implemented, providing the very similar values to the frequency domain technique.

B. Generation and Evaluation of Manoeuvres

The flight data available for the Ranger2000 is used for two purposes in this paper. The predicted aerodynamic forces and moments are validated against the identified model in the usual way. Secondly, the manoeuvres generated by the time optimal prediction code are tested against the flight manoeuvres to show that they are qualitatively similar. This is important so that the time accurate replay of the optimal manoeuvres is for realistic motions. To generate the optimal manoeuvres the following assumptions and approach is taken.

- The propulsive moment terms are zero in the aircraft equations of motion.
- The variation of the engine force with altitude and flight speed is estimated using semi-empirical equations given in Roskam.³⁶
- No ground trajectory is reported in the test data, so these values are defined by an upper and lower value.
- Moments of inertia are not given for each manoeuvre. The maximum take-off data are used for all cases.
- The vertical movement of the centre of gravity is not available and therefore, not considered.

Three manoeuvres are generated. The first is the lazy eight (Fig. 9-a). The aircraft performs a climbing and rolling followed by a diving turn until the final aircraft heading is 180° changed from the initial heading.

A manoeuvre is defined in DIDO using the available experimental tables. The aircraft is allowed to roll from 0° to 90° . The manoeuvre is initiated from a straight and wing level position, then starts a climb while increasing the roll and yaw angles at the same time. Near the maximum altitude, the roll angle has reached 90° and then the aircraft starts to dive with rolling towards the left wing. The heading is still increasing until it reaches 180° and the aircraft is back to the initial altitude, velocity and the roll angle.

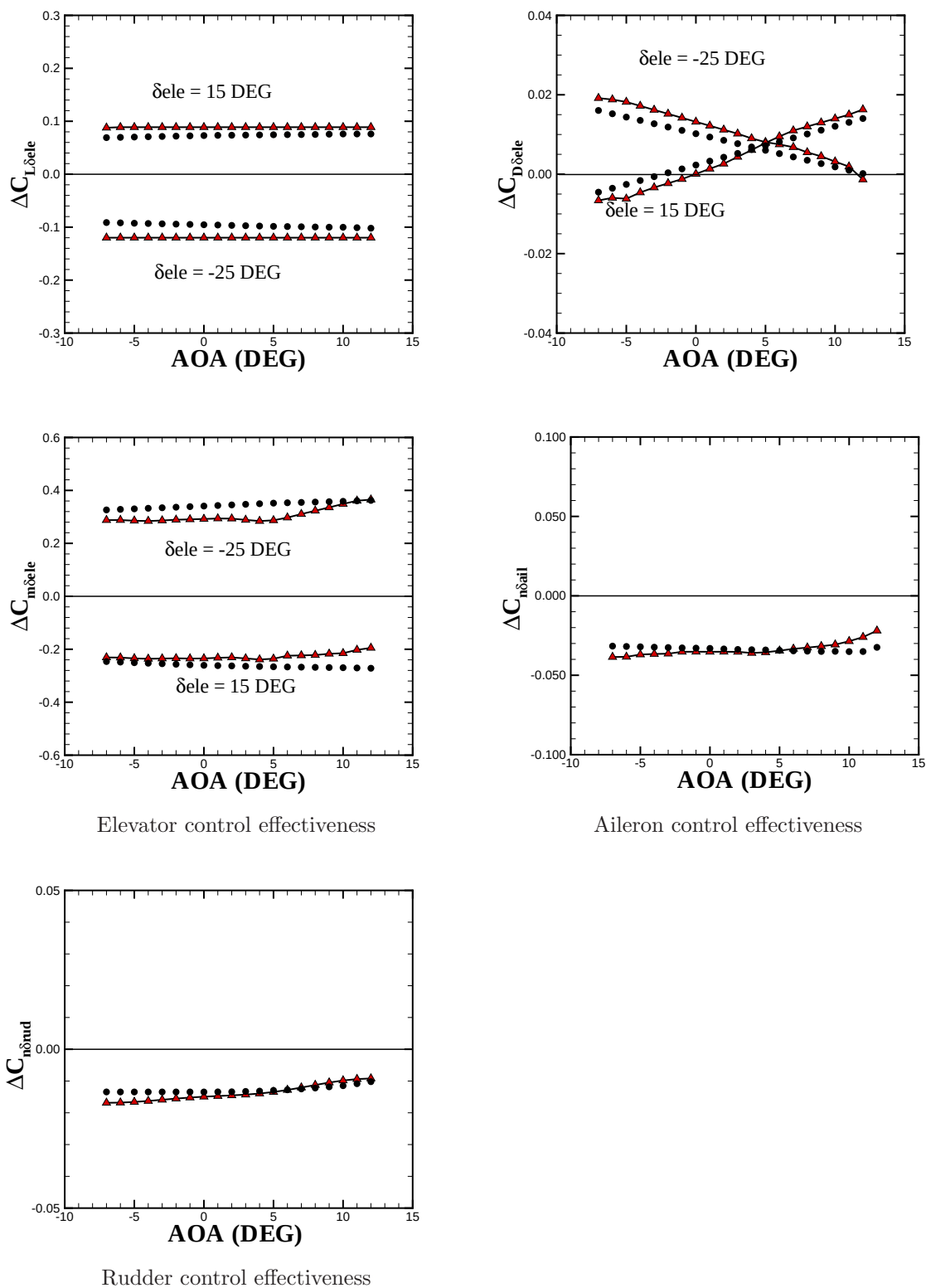


Figure 8. Control tables at M=0.25. The deflection angle for aileron and rudder is nine degrees. The circles denote experiments.

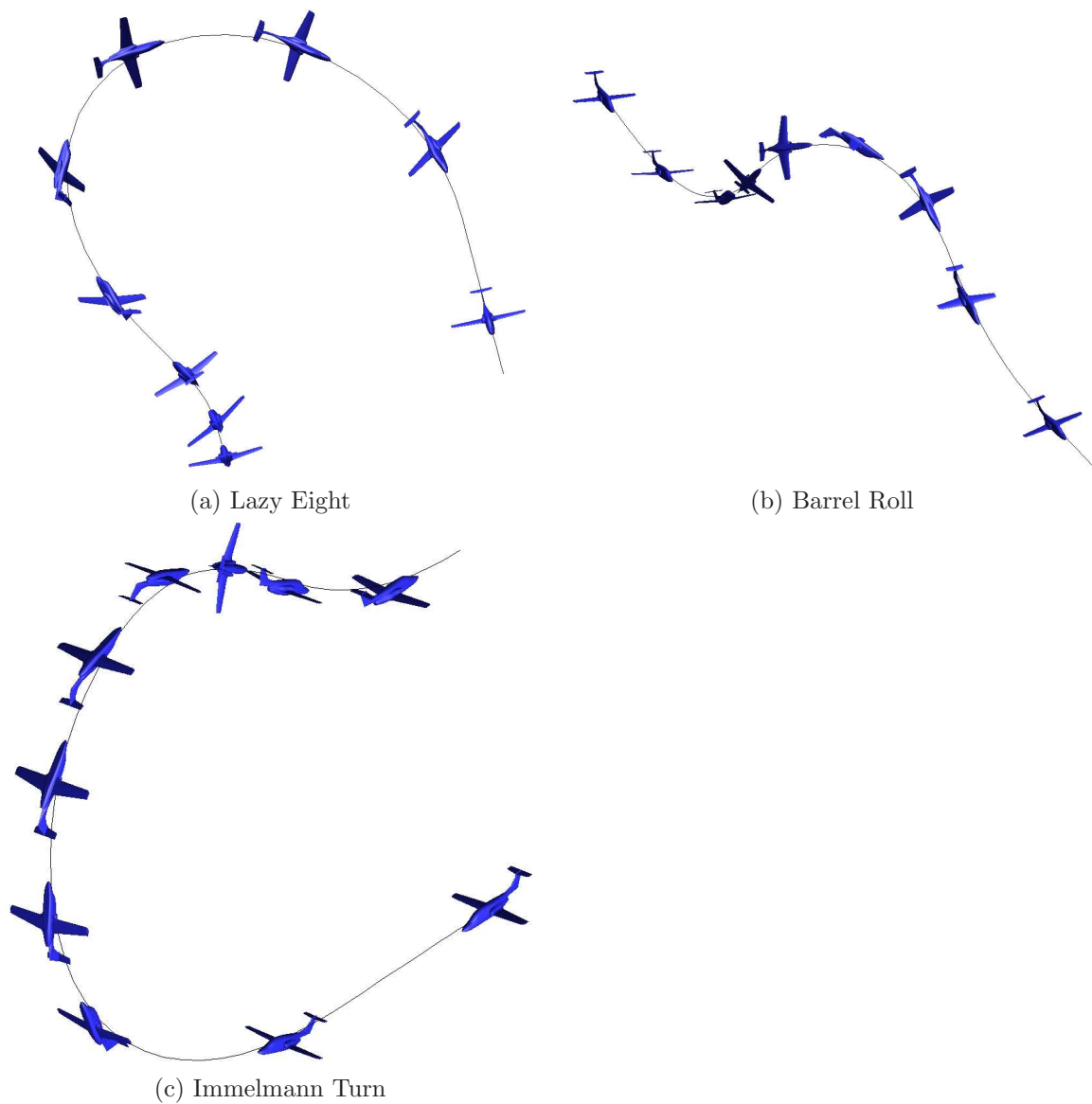


Figure 9. Manoeuvres generated using DIDO and experimental aero data. The aircraft to the flight path scale in the barrel roll and lazy eight is 100:1. For the Immelmann Turn this is 50:1

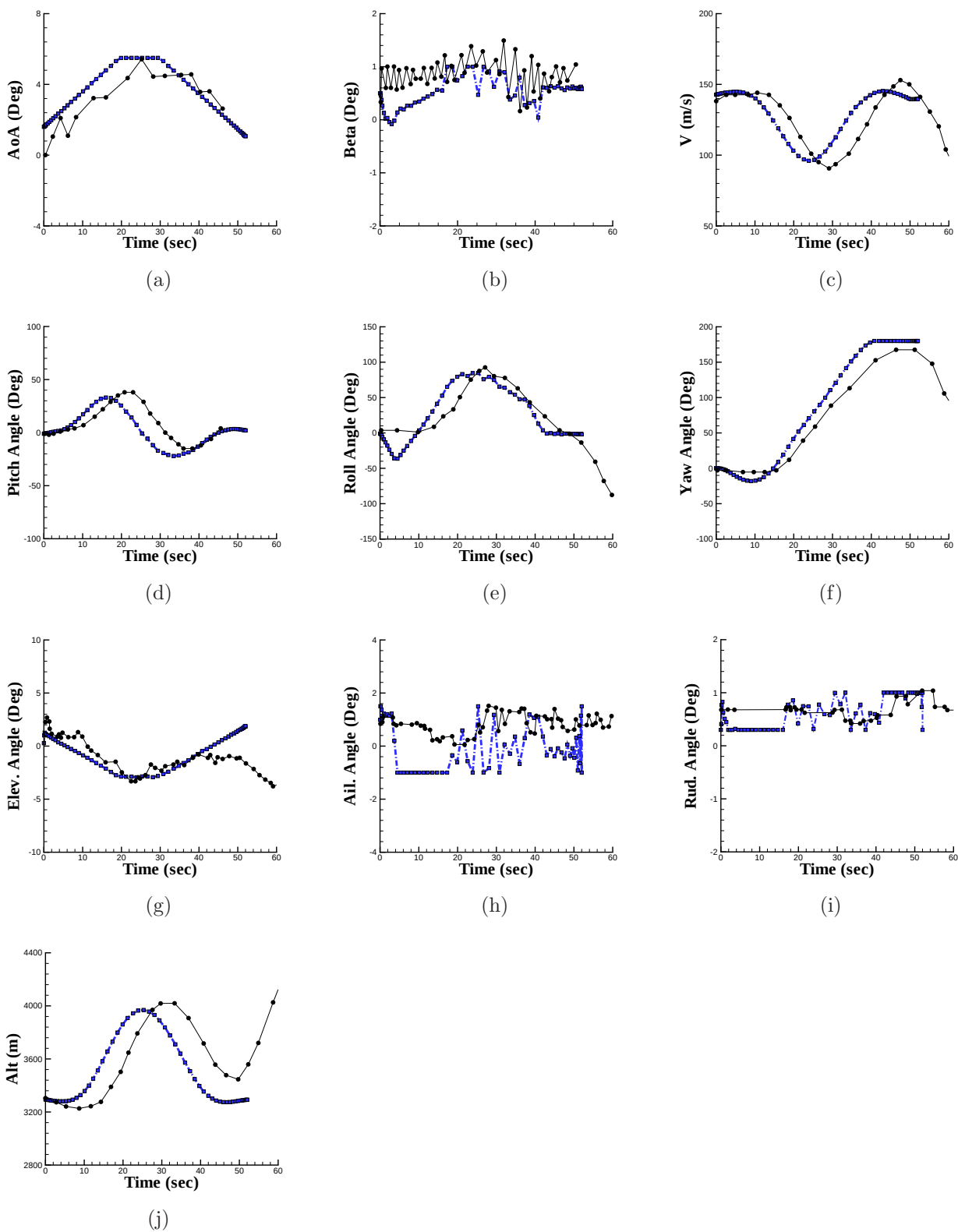


Figure 10. Time histories of states and controls of Lazy eight simulated in DIDO. The results correspond to 60 DIDO nodes, using $CG=24.077\%$ of mean aerodynamic chord and the aircraft mass of 3498 kg. The lines with filled circles show the flight test data.

An optimal solution is found and the results are shown against flight test data in Fig. 10. The simulation results closely follow the trend of flight test data.

Next, for a barrel roll (Fig. 9-b) the aircraft performs a complete rotation around its longitudinal axis. For a barrel roll to the left, the manoeuvre is initiated by a pitch up and a roll turn to the left. During this part of the flight, the left wing is the lowest wing, while the aircraft yaws to the left. At the maximum altitude, the aircraft is nearly upside down. During the second half of the roll, the right wing is lowest and the aircraft tends to yaw to the right.

Again DIDO was used to re-generate the manoeuvre. The roll test angles are in the range of $[-180^\circ, 180^\circ]$. The aircraft is initially straight and wing level, then starts a roll turn, while climbing at the same time. Near the maximum altitude, the roll angle has reached -180° , i.e. the aircraft is upside down. Next, the aileron is gradually relaxed to complete the turn by bringing the roll angle again to zero.

The starting speed and altitude of the minimum time manoeuvre are set to 158 ms^{-1} and 6076 m, respectively. Also, the angles of attack should not exceed the upper limit of 12 degrees. The elevator, aileron and rudder deflection values are also limited to the upper and lower values given in flight test data. An optimal solution was found and the results are shown against flight test data in Fig. 11. The results show that the optimal solution resembles the flight test data. The main differences are probably due to setting take-off moments of inertia for this manoeuvre.

The Immelmann turn, a combined roll and loop manoeuvre, is initiated with a pitch up flight at the entry, and then continues to pull back on the controls such that the aircraft draws a loop in the longitudinal plane. The rudder and ailerons are deflected in a way that the half-loop becomes straight when viewed from the ground. At the end of this loop, the aircraft is upside down, then the aileron(s) is applied to execute a half-roll turn to correct the aircraft orientation. The manoeuvre is terminated at a higher altitude and a 180 degrees reversion of heading. The initial flight speed plays an essential role to accomplish this manoeuvre.

Based on available flight test data, an Immelmann turn manoeuvre is defined. The simulated manoeuvre changes the heading of the aircraft from 0 to 180° , while the final values of the velocity and latitude (x) are fixed with those at the initial time. The starting speed and altitude of the minimum time manoeuvre are set to 163.57 ms^{-1} and 2986m, respectively. Also the angles of attack should not exceed the upper limit of 12 degrees. The elevator, aileron and rudder deflection values are also limited to the upper and lower values given in flight test data. The final manoeuvre is shown in Fig. 12 plotted against flight test data.

The results show that predicted values of altitude, aircraft flight speed and Euler angles match well with flight test data. Although the general trends of angle of attack and elevator deflection are also closely captured using DIDO time-optimal solution, the elevator oscillates at initial steps and there is a discrepancy at final values. High frequency values are seen for the aileron and rudder deflections.

The Immelmann turn contains an inherent discontinuity point. During the upward trajectory and before the pitch angle of the aircraft hits the maximum allowed angle of 90° , the heading and bank angles remain virtually constant. Just after the maximum pitch angle point, the aircraft orientation is suddenly reversed, resulting in a sharp jump of the heading and bank angle values at this point. The presence of discontinuities incurs serious theoretical and numerical difficulties for the optimal control problem.^{7,7} The implementation of standard direct pseudospectral(PS) method for the trajectory optimization problems with the jump discontinuities in the states exhibit an over-shoot of predictions at the jump, named Gibbs phenomenon. In such a case, the nodes refinement might lead to inefficiencies and an ill-conditioned problem.

The step discontinuity is seen around $t = 20$ seconds. The lateral predictions are influenced by the presence of this discontinuity. Very small amplitude Gibbs oscillations are recorded at the yaw and roll angles, however the impacts are much larger for predicted values of side-slip angle and surface deflection of rudder and aileron.

C. Replay of Manoeuvres

The unsteady Euler solver was applied to the DIDO solution in order to evaluate direct aerodynamic coefficients. The predictions were compared against aerodynamic forces from the tabular model. The comparisons test the CFD formulation of the manoeuvre replay, which is done in a time accurate fashion with control surface deflections. The lateral force and moment coefficients are small for the studied manoeuvres, here, only longitudinal forces and moment are compared.

In all cases, the full aircraft geometry was used with the grid scaling of 1.0. A converged steady-state solution was used as the starting condition. These conditions correspond to the initial point ($t = 0$) of each manoeuvre. The physical time step correspond to the number of nodes used in the optimal code, DIDO.

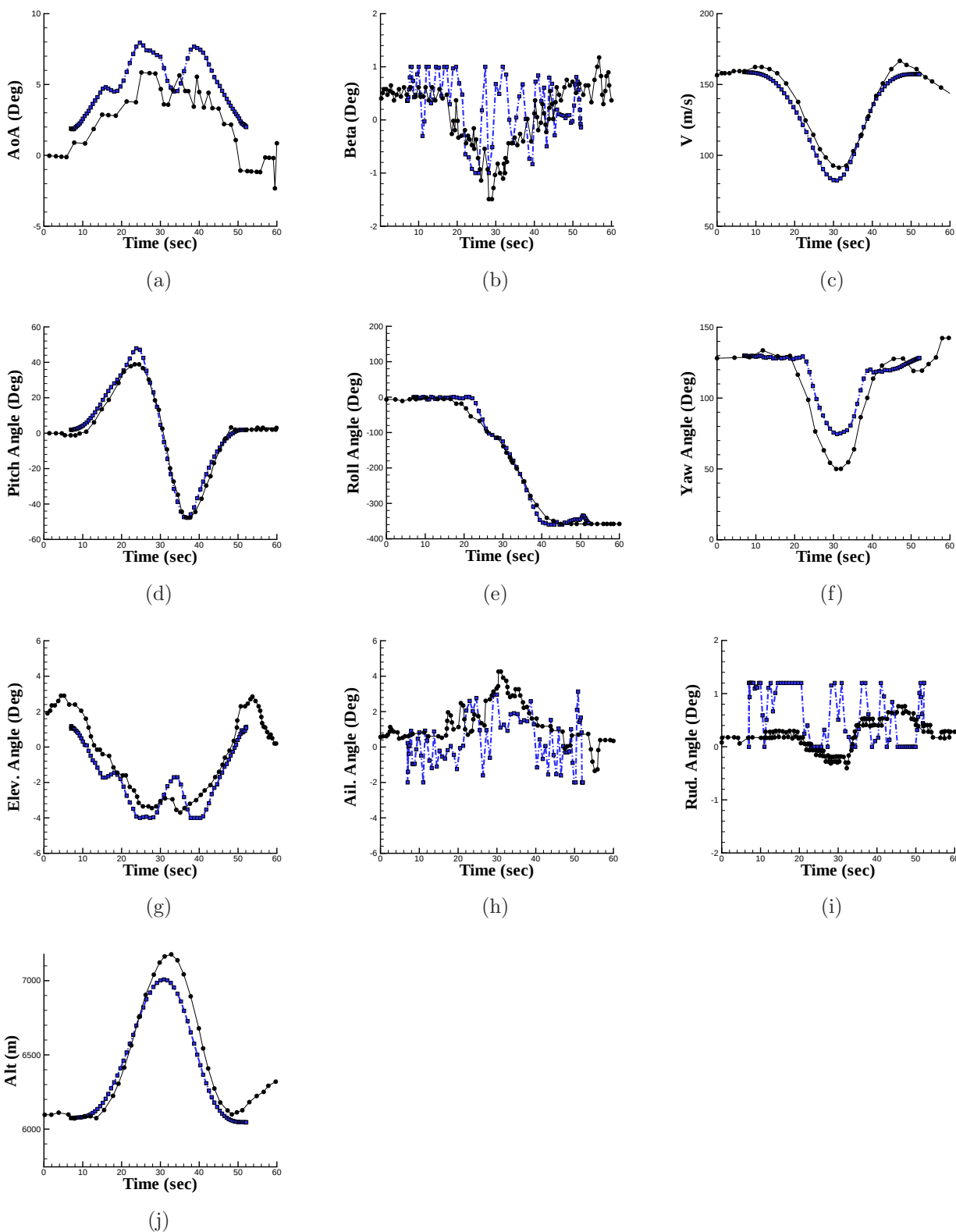


Figure 11. Time histories of states and controls of Barrel roll simulated in DIDO. The results correspond to 60 DIDO nodes, using $CG=24.297\%$ of mean aerodynamic chord and the aircraft mass of 3653 kg. The lines with filled circles show the flight test data.

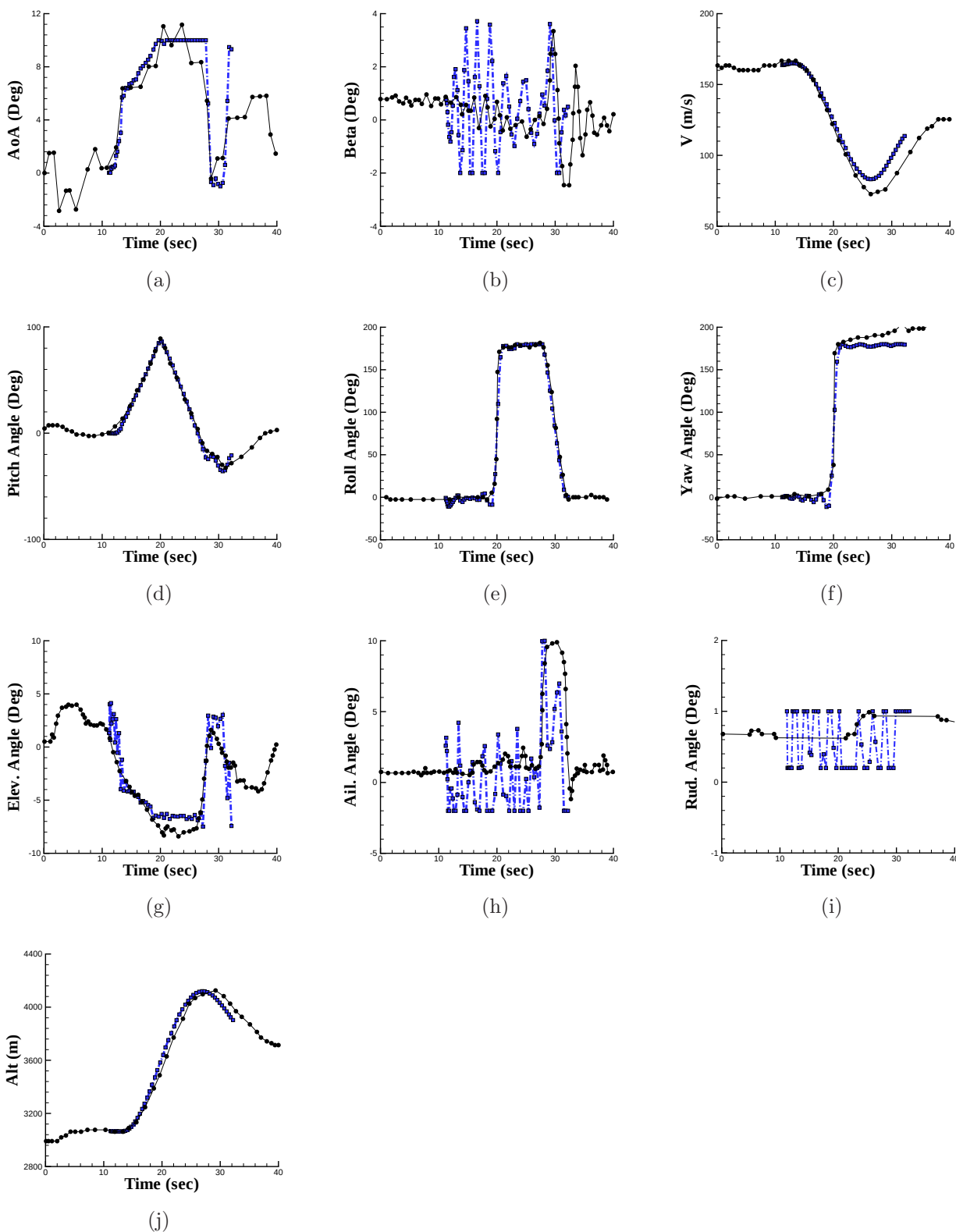


Figure 12. Time histories of states and controls of Immelmann Turn simulated in DIDO. The results correspond to 60 DIDO nodes, using $CG=24.265\%$ of mean aerodynamic chord and the aircraft mass of 3629 kg. The lines with filled circles show the flight test data.

The CFD solver time is found from scaling the physical time based on the grid reference length and the reference velocity. The number of the time step was set to $n_s = 5000$ that has a uniform stepping from t_1 to t_{n_s} . The aircraft reference point velocity at each time step was defined from the initial conditions and the interpolation of DIDO solution. The surface-volume solutions and the body-axis forces and moments computed at each CFD time step are available.

The first manoeuvre presented is the Lazy-Eight. Based on the time histories of the DIDO solution, the values of aircraft reference point velocity, $\mathbf{v}_a$ at each time step was defined. The initial steady state velocity, $\mathbf{v}_0$, was set to 143 m/s. The steady state solution has 1.6 degree angle of attack obtained from the initial point of the manoeuvre. The tabular and replay values for the lift, drag and pitching moment are compared in Fig. 13. The results show that the tables are in perfect agreement with the replay solution. During the manoeuvre the angle of attack remains below six degrees with a maximum Mach number of 0.44. This aircraft has no time hysteresis at such a conditions and as expected the tables perfectly match the direct solution from the replay.

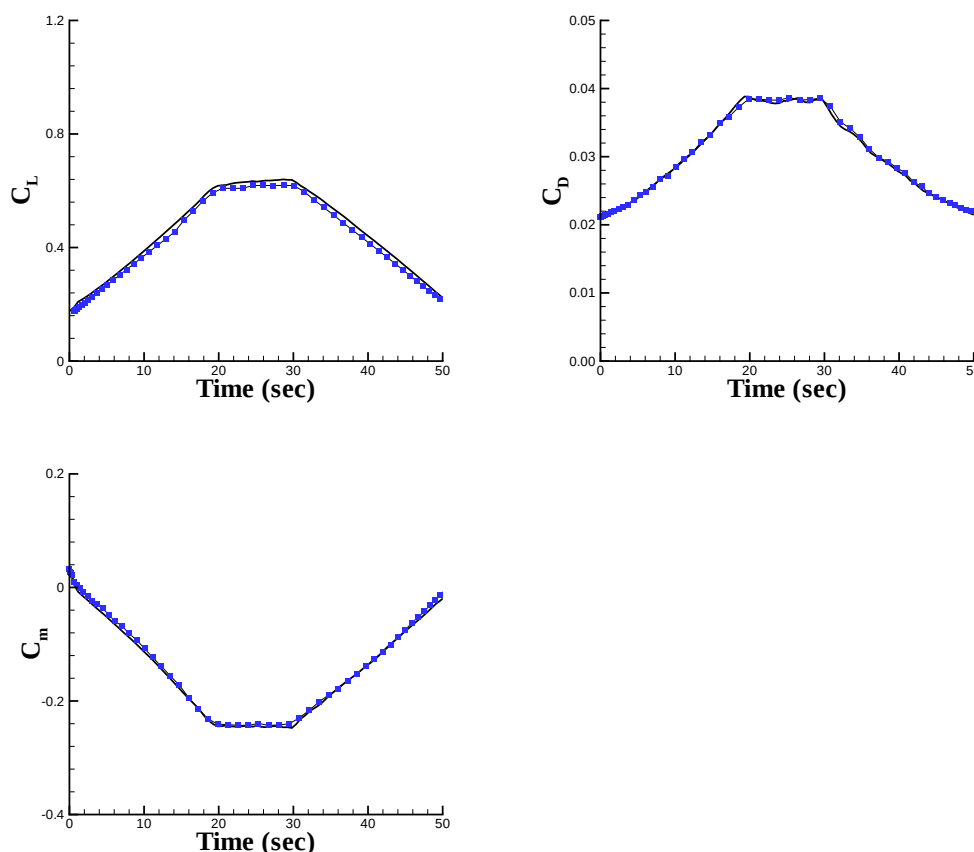


Figure 13. Lazy-Eight replay solution. The solid line shows the replay simulation.

The second manoeuvre considered is the Barrel-roll. The initial steady conditions are: $\mathbf{v}_0 = 159$ m/s and an angle of attack of 1.9 degrees. From the DIDO solution, the grid point locations and grid point velocities are set. The tabular and replay values for the lift, drag and pitching moment are compared in Fig. 14. The results show good agreement with the replay solution. The angle of attack during manoeuvre time remains below eight degrees, while a good agreement is expected between the tables and the replay.

Finally, the results of the Immelmann turn are presented in Fig. 15. The initial steady conditions are: $\mathbf{v}_0 = 164$ m/s and AOA=0.06. The trends of lift, drag and pitching moment closely follow the replay with some localized differences between the replay and tabular values. There are no history effects from the state parameters since the aircraft flies in aerodynamically benign regions of the flight envelope. The differences are mainly due to dynamic effects from the very rapid motion of the control surfaces. As discussed earlier, the Immelmann turn contains an inherent discontinuity point and results in the time optimal solution having

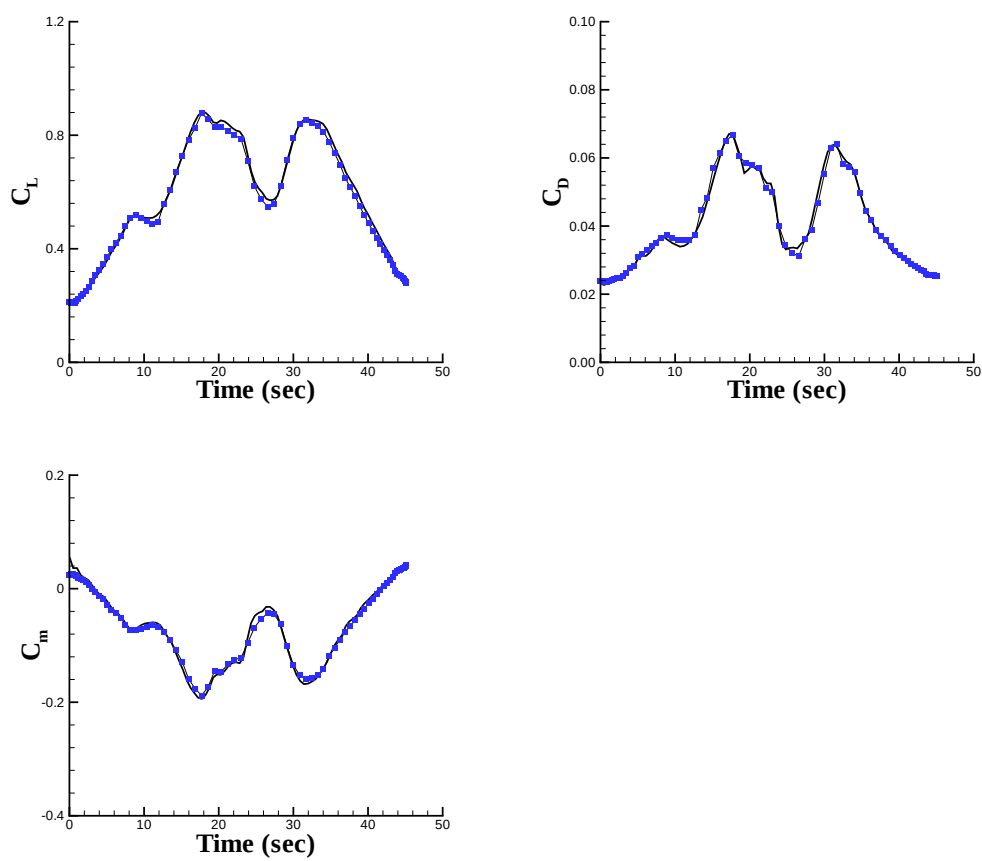


Figure 14. Barrel Roll replay solution. The solid line shows the replay simulation.

high frequency and large amplitude motion of the control surfaces. The effects of rapid motion of the control surface have been discussed in a paper by Ghoreyshi et al.²⁰

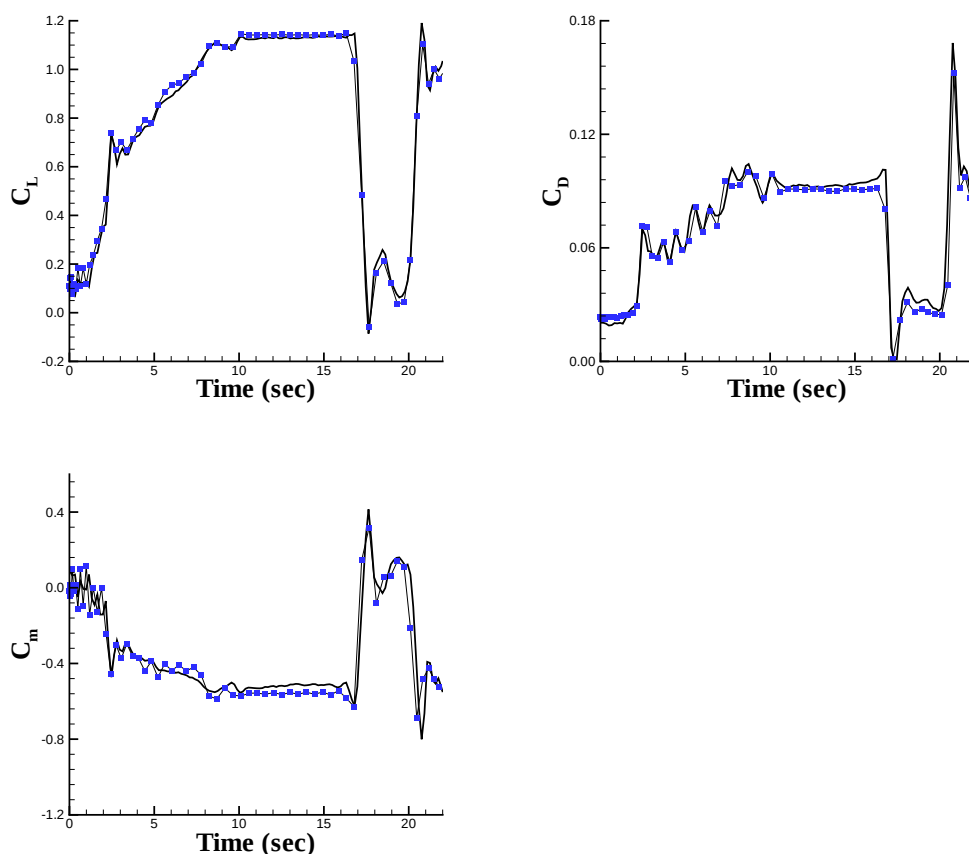


Figure 15. The Immelmann turn replay solution. The solid line shows the replay simulation.

V. Conclusions

The paper investigate the validity of the tabular approach for the maneuvering aircraft by testing the aerodynamic model through replaying manoeuvres using an unsteady CFD calculation to check the consistency of the aerodynamic forces and moments. The CFD solver uses two types of mesh movement, namely rigid motion, and transfinite interpolation for the control surfaces.

The Euler equations were used to demonstrate the replay framework. Sampling and Data fusion was used to allow the generation of the aerodynamic tables in a feasible number of CFD calculations. The validation against available experimental data showed the solutions are credible for the shock effects, but they fail to predict the drag force and lateral coefficients due to lack of skin friction.

Three aerobatic manoeuvres were generated using the time-optimal solver, DIDO. The results were compared with available flight test data. The trajectories of both simulation and test data match well for all cases. In terms of control values, the Immelmann turn simulation includes high frequency motion of the surface deflections, due to the presence of a discontinuity point in the roll angle. The time-optimal solver for the trajectory optimization experiences problems with the jump discontinuities in the states due to an over-shoot of predictions at the jump from the Gibbs phenomenon. This results in the formation of spikes in the replay solutions due to dynamic effects from the very rapid motion of the control surfaces.

The results of tabular predictions were compared against the coefficients from the replaying the manoeuvre using an unsteady CFD calculation. For all considered manoeuvres the aircraft flies in aerodynamically benign regions of the flight envelope and the tables perfectly match the replay solution. The results of

this paper demonstrated the validity of using tables for such conditions. Overall, the paper illustrates the validity of the CFD framework for generating aerodynamic tables and replaying manoeuvres to test them. This framework, when applied to more demanding manoeuvres can show the limitations in the tabular formulation.

VI. Acknowledgements

Computer time for RANS calculations was provided through the UK Applied Aerodynamics Consortium (UKAAC) under Engineering and Physical Sciences Research Council grant EP/F005954/1. This work was partly done under funding from the European Union sixth framework project SimSAC (Simulating Stability and Control).

References

- ¹Bryan, G. H. and Williams, W. E., "The Longitudinal Stability Of Aerial Gliders", 1904, Proceeeding of Royal Society, London, Vol. 33, Series A., pp. 100-116
- ²Greenwell, D. I., "A Review of Unsteady Aerodynamic Modelling for Flight Dynamics of Manoeuvrable Aircraft", 2004, AIAA 2004-5276, AIAA Atmospheric Flight Mechanics Conference and Exhibit, 16 - 19 August, Providence, Rhode Islandm
- ³Hamel, P. G. and Jategaonkar, R. V., Evolution of Flight Vehicle System Identification, Journal of Aircraft, Vol. 33, No. 1, 1996, pp. 9-28
- ⁴Jouannet, C. and Krus, P., Lift Coefficient Predictions for Delta Wing Under Pitching Motions, 32nd AIAA Fluid Dynamics Conference and Exhibit AIAA-2002-2969, 24-26 June 2002, St. Louis, Missouri
- ⁵Katz, J. and Schiff, L. B., "Modeling Aerodynamic Responses to Aircraft Maneuvers A Numerical Validation", Journal of Aircraft, 1986, Vol. 23, No. 1, pp. 19-25
- ⁶Tobak, M. and Schiff, L.B., Aerodynamic Mathematical Modeling: Basic Concepts, AGARD Lecture Series on Dynamic Stability Parameters, Lecture No. 1, March 1981.
- ⁷Tobak, M., Chapman, G. T., and Schiff, L. B., "Mathematical Modeling of the Aerodynamic Characteristics in Flight Dynamics", 1984, NASA technical report, TM 85880.
- ⁸Goman, M. and Khrabrov, A., "State-Space Representation of Aerodynamic Characteristics of an Aircraft at High Angles of Attack", 1992, AIAA Paper 92-4651-CP.
- ⁹Klein, V. and Nodere, K. D., "Modeling of Aircraft Unsteady Aerodynamic Characteristics", 1994, NASA technical report, National Aeronautics and Space Administration Langley Research Center Hampton, Virginia, 23681-0001
- ¹⁰Rudnik, R. and Germain, E., "Re-No. Scaling Effects on the EUROLIFT High Lift Configurations", 2007, 45th AIAA Aerospace Sciences Meeting and Exhibit, AIAA 2007-752, January 2007, Reno, Nevada
- ¹¹Jirasek, A., Jeans, T. L., Martenson, M., Cummings, R. M. and Bergeron, K., "Improved Methodologies for Maneuver Design of Aircraft Stability and Control Simulations", 2010, 48th AIAA Aerospace Sciences Meeting Including the New Horizons Forum and Aerospace Exposition, AIAA 2010-515, January 2010, Orlando, Florida
- ¹²Da Ronch, A., Valespin, D., Ghoreyshi, M. and Badcock, K. J., "Computation and Evaluation of Dynamic Derivatives using CFD", 2010, AIAA-2010-4817 28th AIAA Applied Aerodynamics Conference, Chicago, Illinois, June 28-1
- ¹³Da Ronch, A., Ghoreyshi, M., Badcock, K. J., Gortz, S., Widhalm, M. and Dwight, R. "Linear Frequency Domain and Harmonic Balance Predictions of Dynamic Derivatives", 2010, AIAA-2010-4699, 28th AIAA Applied Aerodynamics Conference, Chicago, Illinois, June 28-1.
- ¹⁴Ballhaus, W.F. and Goorjian, P.M., Computation of Unsteady Transonic Flow by the Indicial Method, AIAA Journal, Vol. 16, 1978, pp. 117-124.
- ¹⁵Rizzetta, D.P., Time-Dependent Response of a Two-Dimensional Airfoil in Transonic Flow, AIAA Journal, Vol. 17, 1979, pp. 26-32.
- ¹⁶Chyu, W.J. and Schiff, L.B., Nonlinear Aerodynamic Modeling of Flap Oscillations in Transonic Flow: A Numerical Validation, AIAA Journal, Vol. 21, 1983, pp. 106-113.
- ¹⁷Steger, J.L. and Bailey, H.E., Calculation of Transonic Aileron Buzz, AIAA Journal, Vol. 18, 1980, pp. 249-255.
- ¹⁸Allan, M.R., Badcock, K. J. and Richards, B. E., CFD Based Simulation of Longitudinal Flight Mechanics With Control. 43rd AIAA Aerospace Science Meeting and Exhibition, AIAA-2005-46.
- ¹⁹Schutte, A., Einarsson, G., Raichle, A., Schoning, B., Mnnich, W. and Forkert, T., Numerical Simulation of Manoeuvring Aircraft by Aerodynamic, Flight Mechanics, and Structural Mechanics Coupling, Journal of Aircraft Vol. 46, No. 1, 2009, pp. 53 -64
- ²⁰Ghoreyshi, M., Badcock, K., Da Ronch, A., Swift, A., Marques, S. and Ames, N., "Framework for Establishing the Limits of Tabular Aerodynamic Models for Flight Dynamics Analysis", 2009, AIAA-2009-6273, AIAA Guidance, Navigation, and Control Conference, Chicago, Illinois, Aug. 10-13.
- ²¹Ghoreyshi, M., Badcock, K. J. and Woodgate, M., Accelerating the Numerical Generation of Aerodynamic Models for Flight Simulation, Journal of Aircraft, Vol. 46, No. 3, pp. 972- 980
- ²²Badcock, K.J., Richards, B.E. and Woodgate, M.A. Elements of Computational Fluid Dynamics on Block Structured Grids using Implicit Solvers. Progress in Aerospace Sciences, vol 36, 2000, pp 351-392.
- ²³Boelens, O.J., Badcock, K.J., Elmilgui, A., Abdol-Hamid, K.S. and Massey, S.J., Comparison of Measured and Block Structured Simulation Results for the F-16XL Aircraft, Journal of Aircraft, 46(2), March, 2009, 377-384.

- ²⁴von Kaenel, R., Rizzi, A., Oppelstrup, J., Goetzendorf-Grabowski, T., Ghoreyshi, M., Cavagna, L. and Berard, A., "CEASIOM: Simulating Stability & Control with CFD/CSM in Aircraft Conceptual Design", 2008, 26th International Congress of the Aeronautical Sciences, ICAS.
- ²⁵Betts, J. T., Survey of Numerical Methods for Trajectory Optimization, 1998, Journal of Guidance, Control, and Dynamics, Vol. 21, No. 2, pp. 193-207.
- ²⁶Hull, D. G., Conversion of Optimal Control Problems into Parameter Optimization Problems, 1997, Journal of Guidance, Control, and Dynamics, Vol. 20, No. 1, pp. 5760.
- ²⁷Brinkman, K. and Visser, H. G., Optimal turn-back manoeuvre after engine failure in a single-engine aircraft during climb-out, Journal of Aerospace Engineering, Vol. 22, Part G., 2006, pp. 17-27.
- ²⁸Ross, I. M., and Fahroo, F., "Users Manual for DIDO 2002: A MATLAB Application Package for Dynamic Optimization," 2002, NPS Technical Rept, Dept. of Aeronautics and Astronautics, Naval Postgraduate School, Monterey, CA, AA-02-002.
- ²⁹The Mathworks, Inc., MATLAB version 2008a, Inc. <http://www.mathworks.com>, 2008.
- ³⁰Gottlieb, D., Hussaini, M. Y., and Orszag, S. A., Theory and Applications of Spectral Methods, 1984, Spectral Methods for PDEs, Society for Industrial and Applied Mathematics, Philadelphia.
- ³¹Canuto, C., Hussaini, M. Y., Quarteroni, A., and Zang, T. A., "Spectral Methods in Fluid Dynamics", 1988, Springer-Verlag, New York, 1988.
- ³²Elnagar, J., Kazemi, M. A., and Razzaghi, M., The Pseudospectral Legendre Method for Discretizing Optimal Control Problems, 1995, IEEE Transactions on Automatic Control, Vol. 40, No. 10, pp. 1793-1796.
- ³³Etkin, B., Dynamics of Flight, Stability and Control, 2nd Edition, Wiley, 1982.
- ³⁴Steven, B.L. and Lewis, F. L., Aircraft Control and Simulation, Wiley, 1992.
- ³⁵Rampurawala, A.M. and Badcock, K.J., Evaluation of a Simplified Grid Treatment for Oscillating Trailing-Edge Control Surfaces, Journal of Aircraft, 44(4), 2007, 1177-1188.
- ³⁶Roskam, J., "Airplane Design. Part I through VIII", 1990, Roskam Aviation and Engineering Corporation, Kansas, USA.
- ³⁷Homescu, C. and Navon, I. M., "Optimal control of flow with discontinuities", 2003, Journal of Computational Physics, Vol. 187 pp. 660682.
- ³⁸Ross, I. M., Fahroo, F., "Pseudospectral Knotting Methods for Solving Optimal Control Problem", 2004, Journal of Guidance, Control, and Dynamics Vol. 27, No. 3, pp. 397-405.